Persistence Beyond Productivity: A Process-Tracing Analysis of the Butterfly Net Fishery in Lake Pátzcuaro, Mexico

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Verano Internacional de la Ciencia · REDNACECYT-EAAMO Bridges 2026

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1 Context & Motivation

By economic logic, the "butterfly net" should have disappeared by now.

The net is historically tied to fishing the **t'iru** (*Goodea atripinnis*), a species now nearly extinct (*West, 1948*). Changing fishing conditions and the arrival of other gear transformed the context in which this tool was used.

Yet the net is still on the water — and in Janitzio it earns more than ever: it has become a symbol of cultural representation and a tourist exhibit *(Jolly, 2018)*.

2 Research Question

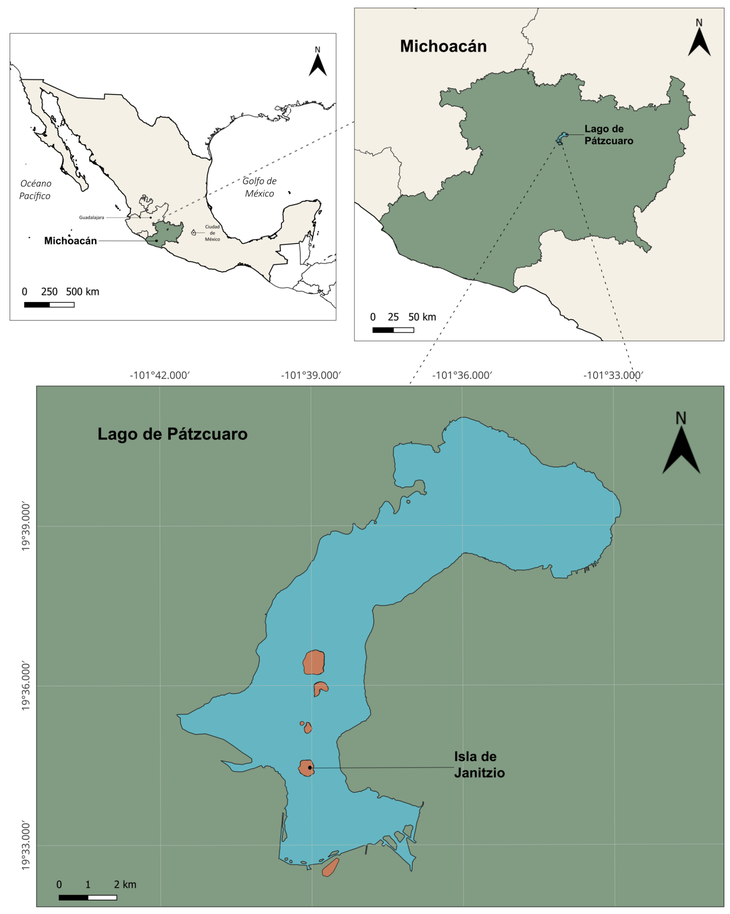
Why does the traditional "butterfly net" fishery persist even though its productive function has weakened?

Hypothesis

The **decline of the t'iru** weakened the net's function as a fishing tool for this species, but did not make it disappear. Its persistence is linked to **fishing adaptation** and to its incorporation into **cultural and tourism practices**, mainly in Janitzio.

Study area

Lago de Pátzcuaro, Michoacán, México



Lake Pátzcuaro and Janitzio Island, Michoacán, Mexico

3 Methodology

Process tracing with historical, biological, regulatory and socioeconomic sources

1. **Identify the hypothesis**
2. **Build the timeline**
3. **Map causal relations**
4. **Identify alternative events**
5. **Identify counterfactual outcomes**
6. **Search for evidence: tests**
   - **Straw in the wind** — Adds plausibility, not decisive
   - **Hoop** — Necessary: failing rules it out
   - **Doubly decisive** — Confirms and rules out rivals
   - **Smoking gun** — Sufficient: passing confirms
7. **Repeat for rival hypotheses** — *tourism · fish decline · regulation*
8. **Hypothesis validated** or falsified

4 Results

The net lost its fish, but not its visitors

	Late 1800s	1930s	1940s	1950–1970	1990–2000	2015–2026
Fishing adaptation	Traditional fishery: the net catches t'iru	Predator fish introduced to raise yield and tourism	The t'iru collapses: the net loses its main catch (*West, 1948*); the lake level drops (*Chacón Torres et al., 2026)*	Gillnets gain ground; the butterfly net shifts to charal *(Contreras Jaimes et al., 2022)*	39 butterfly nets still active for charal, mostly in Janitzio *(Orbe-Mendoza et al., 2002)*	Fishing use persists
Culture & tourism	First tourist excursions to the lake	The lake landscape is promoted for tourism *(Jolly, 2018)*	The nets are already an attraction in Janitzio	Tourism expands; by 1979 fishing loses economic value	Exhibitions for tourists continue	NOM-036 (2015) recognizes the net as demonstration and tourism gear *(DOF, 2015)*

COUPLING · JANITZIO

Fishers without their fish + visitors seeking the image + community organization

The net survives as an exhibit, alongside charal fishing and sales

≈70% of participating families' income (Vargas-Herrejón et al., 2022).

INTEGRATED RESULT: the butterfly net did not disappear: it diversified its functions, kept fishing uses and gained cultural and tourism relevance in Janitzio.

5 Conclusions

Economic logic predicts the net's extinction; its persistence suggests that its **value goes beyond its productive function** and responds to culture and identity.

The net **diversified its function**: catching fish and being exhibited to tourists.

Tourism **predates the transformation**; its interaction with changes in fishing contributed to the net's persistence.

Next steps

Simulate socio-ecological scenarios of the falling level and drying of Lake Pátzcuaro in Janitzio.

Explore how the shrinking surface and depth of the lake and the availability of fish affect fishing, tourism income and community livelihoods.

6 Connection to EAAMO

Participation and representation: the net has shifted from a fishing tool to a symbol of Janitzio's cultural and tourism tradition. This invites reflection on how traditional practices persist and acquire new meanings as fishing changes.



Butterfly-net fishers on Lake Pátzcuaro (2026)

References: **Vargas-Herrejón, M., Medina-Nava, M., y Arellanes-Cancino, (2022)**. *La pesca tradicional en el Lago de Pátzcuaro, Michoacán.* Ciencia Pesquera, Vol. 30, Núm. 1-2, pp. 205-206. ***Orbe-Mendoza, A. A., Acevedo-García, J., & Lyons, J. (2002).*** *Lake Pátzcuaro fishery management plan.* Reviews in Fish Biology and Fisheries, *12(2), pp. 207–217.* **West, R. C. (1948)**. *Cultural geography of the modern Tarascan area*. Smithsonian Institution, Institute of Social Anthropology, Publication No. 7, pp. 54. **Williams, E. (2014).** *La gente del agua: Etnoarqueología del modo de vida lacustre en Michoacán*. El Colegio de Michoacán, pp. 101–102. **Jolly, J. (2018).** *Creating Pátzcuaro, creating Mexico: Art, tourism, and nation building under Lázaro Cárdenas*. University of Texas Press, pp. 185–190. **Mejía Argote, M. C., Ávila Carreón, K., & Ángeles González, C. (2025)** Expresión Económica, Núm. 54, pp. 135–160. Diario Oficial de la Federación. (2015). *Norma Oficial Mexicana NOM-036-SAG/PESC-2015, Pesca responsable en el Lago de Pátzcuaro ubicado en el Estado de Michoacán. Especificaciones para el aprovechamiento de los recursos pesqueros*, pp. 3–4. **Contreras Jaimes, B., Camou Guerrero, A., Ayala Orozco, B., & Urquijo Torres, P. S. (2022).** *Historia de las organizaciones pesqueras en la cuenca del lago de Pátzcuaro, Michoacán, 1930–2020.* Región y Sociedad, pp. 12–18. **Chacón Torres, A. et al. (2026).** *Environmental assessment of Lake Patzcuaro: Morphometric and Water Quality Study.* Limnology and Environmental Assessment, pp. 8-14. **Bennett, A., & Checkel, J. T. (Eds.) (2015).** *Process tracing: From metaphor to analytic tool*. Cambridge University Press, pp. 16–21.